

**Assignment 4**

1. Write a program to generate the following sequences and plot the graph-
  - a. Unit impulse (delay, advance)
  - b. Step sequence (delay, advance)
  - c. Combination of unit impulse (delay/advance) and/or step sequence (delay/advance)
2. Write a program to implement the convolution of two sequences. Simulate the convolution sum step by step.
3. Write a program to implement the autocorrelation and cross-correlation of two sequences.
4. Implement a Moving Average Filter. Suppose there is an original uncorrupted signal  $s[n]=2[n(0.9)^n]$ , and noise  $d[n]=\text{rand}(R,1)-0.5$ .
  - a. Plot the signal  $s[n]$  of 50 points
  - b. Plot a random signal  $d[n]$  of same 50 points
  - c. Subtract 0.5 from each random signal  $d[n]$  and plot them
  - d. Sum  $x[n]=s[n]+d[n]$
  - e. Plot the corrupted signal  $x[n]$
  - f. Do the same process for 50 times

```
For n=1:50
    D=rand(R,1)-0.5
    X=s+d
    X1=X1+x
End
X1=X1/50;
```
  - g. Plot the noise removed signal  $X1$

**Assignment Submission Due: 27/08/2026**